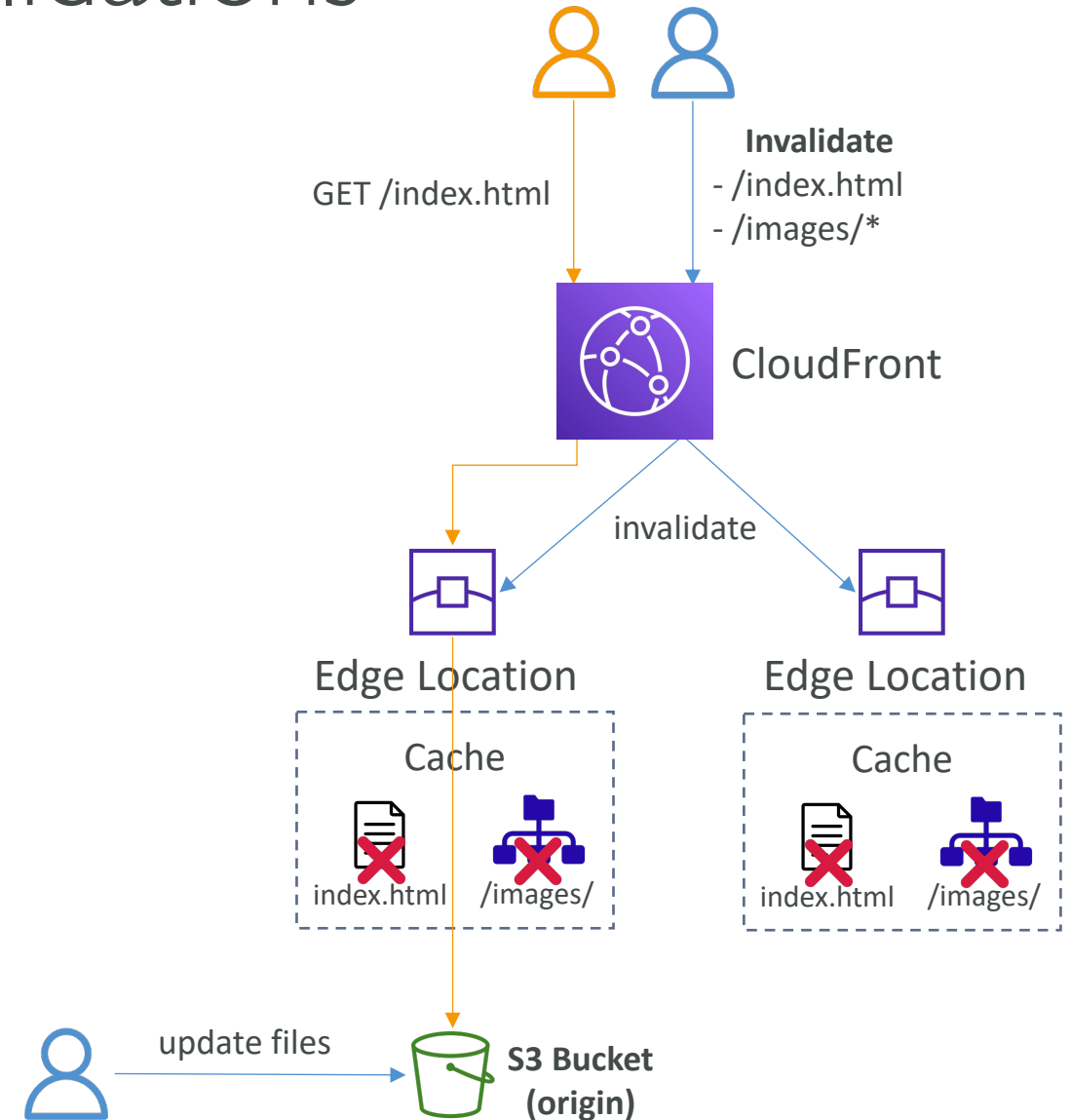


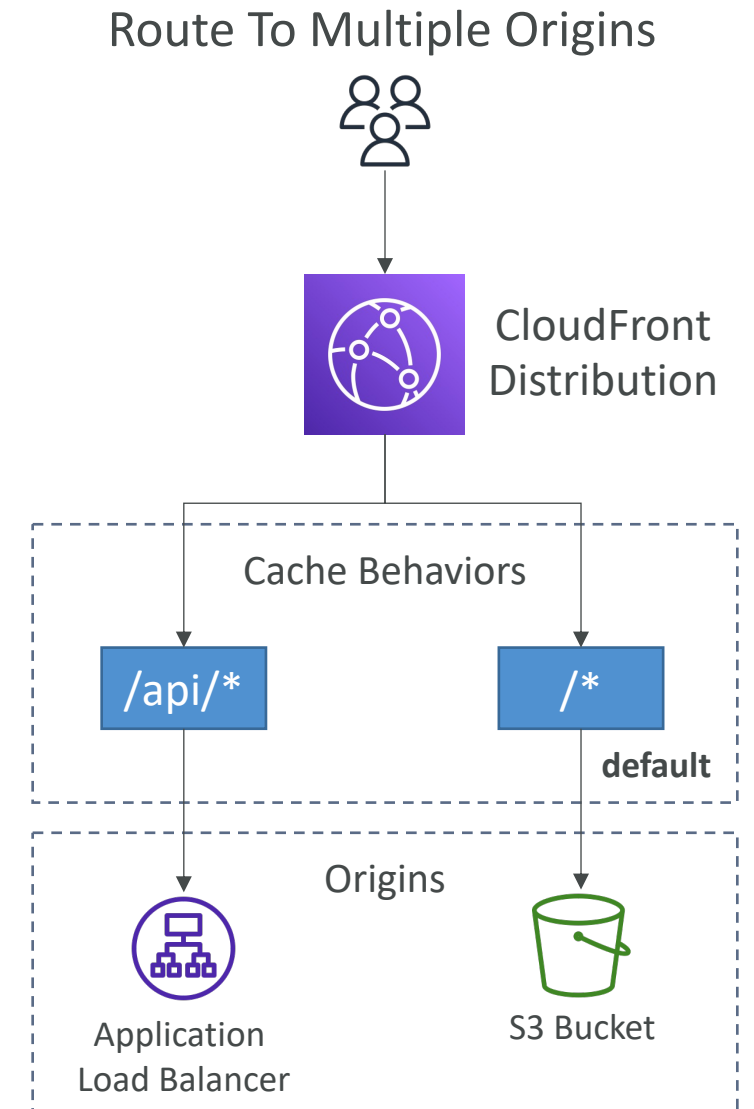
CloudFront – Cache Invalidation

- In case you update the back-end origin, CloudFront doesn't know about it and will only get the refreshed content after the TTL has expired
- However, you can force an entire or partial cache refresh (thus bypassing the TTL) by performing a **CloudFront Invalidation**
- You can invalidate all files (*) or a special path (/images/*)

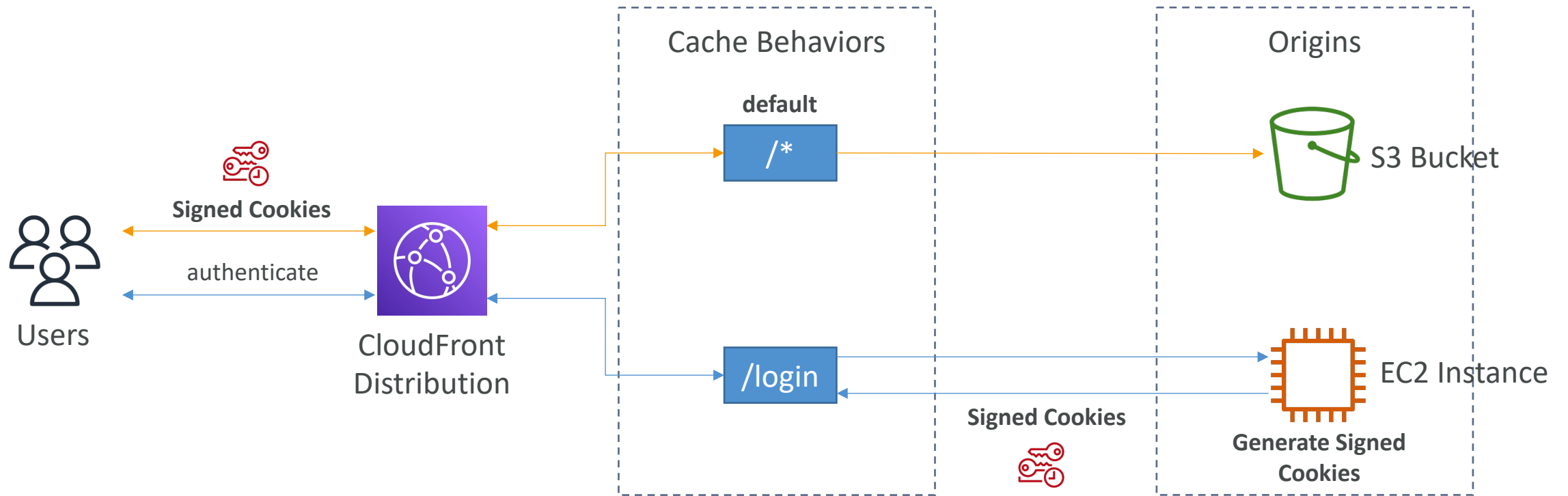


CloudFront – Cache Behaviors

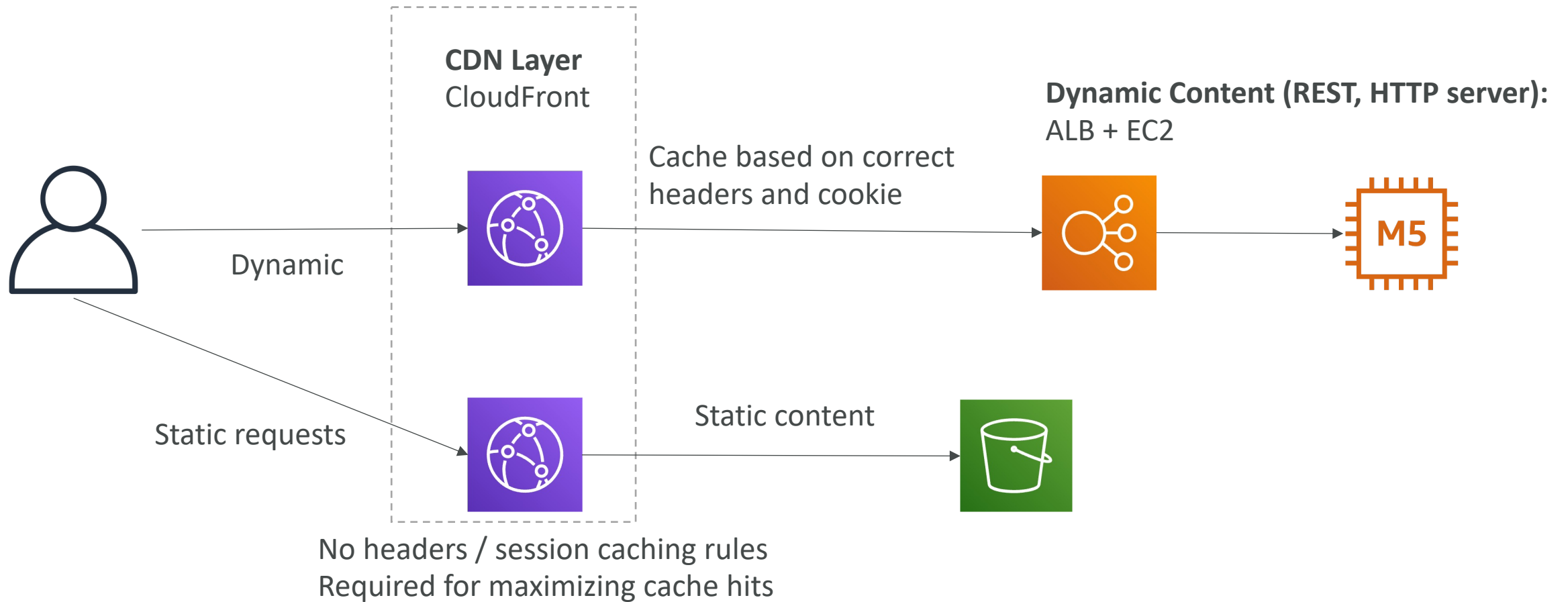
- Configure different settings for a given URL path pattern
- Example: one specific cache behavior to **images/*.jpg** files on your origin web server
- Route to different kind of origins/origin groups based on the content type or path pattern
 - /images/
 - /api/
 - /* (default cache behavior)
- When adding additional Cache Behaviors, the Default Cache Behavior is always the last to be processed **and is always /***



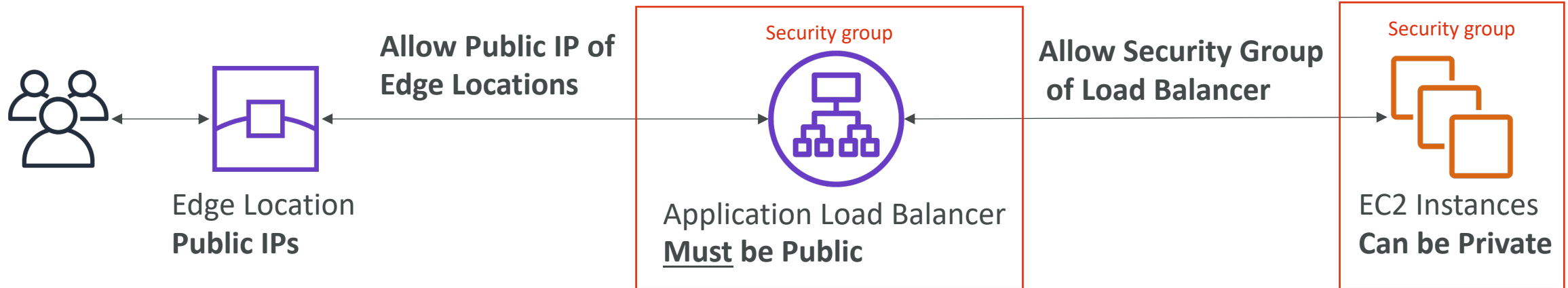
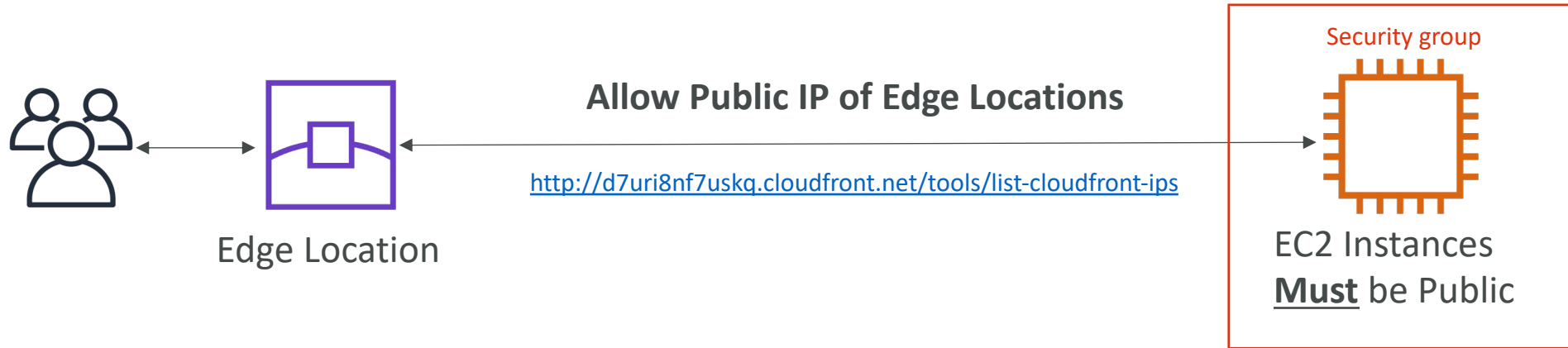
CloudFront – Cache Behaviors – Sign In Page



CloudFront – Maximize cache hits by separating static and dynamic distributions



CloudFront – ALB or EC2 as an origin



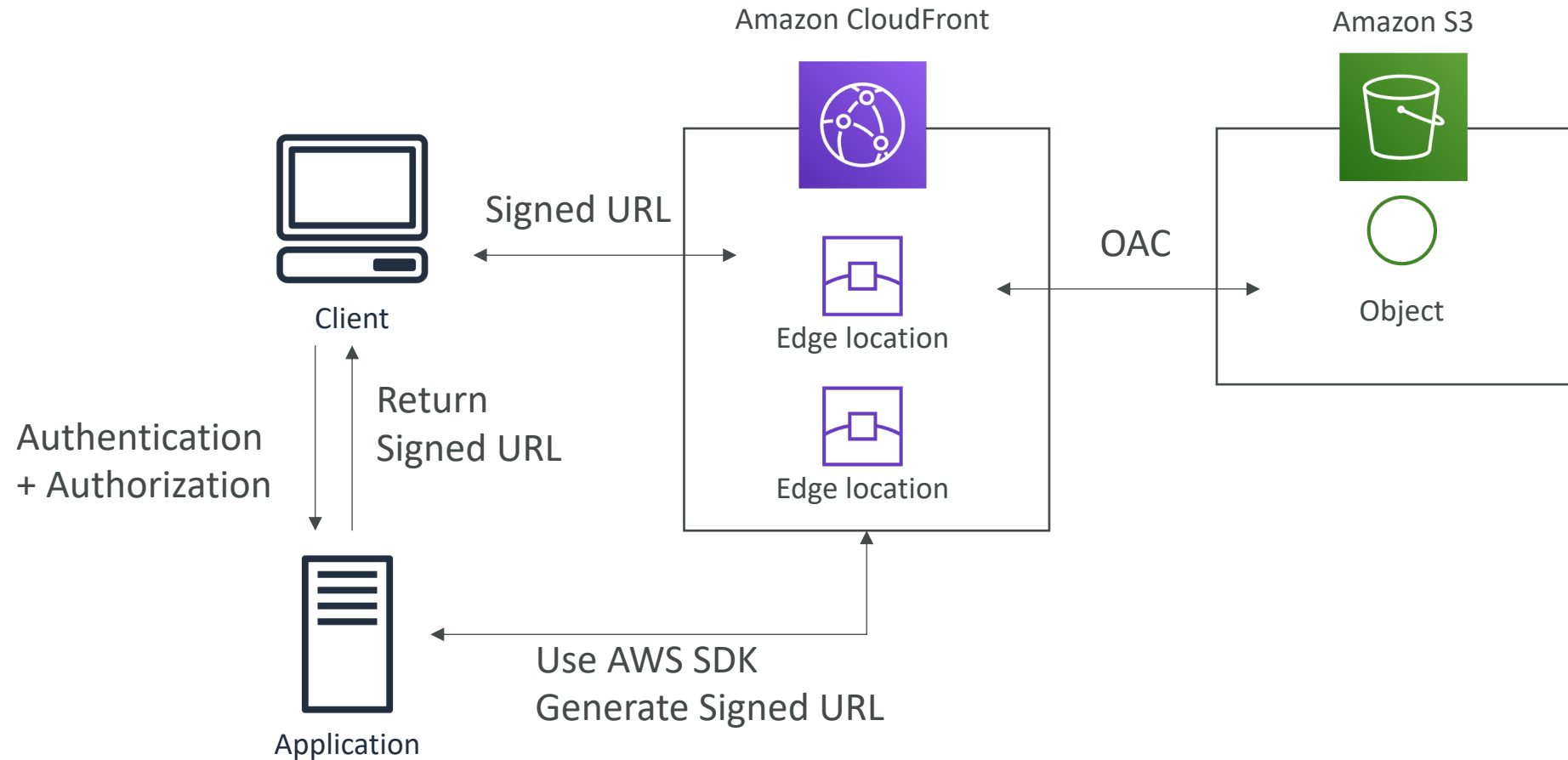
CloudFront Geo Restriction

- You can restrict who can access your distribution
 - **Allowlist:** Allow your users to access your content only if they're in one of the countries on a list of approved countries.
 - **Blocklist:** Prevent your users from accessing your content if they're in one of the countries on a list of banned countries.
- The “country” is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content

CloudFront Signed URL / Signed Cookies

- You want to distribute paid shared content to premium users over the world
- We can use CloudFront Signed URL / Cookie. We attach a policy with:
 - Includes URL expiration
 - Includes IP ranges to access the data from
 - Trusted signers (which AWS accounts can create signed URLs)
- How long should the URL be valid for?
 - Shared content (movie, music): make it short (a few minutes)
 - Private content (private to the user): you can make it last for years
- Signed URL = access to individual files (one signed URL per file)
- Signed Cookies = access to multiple files (one signed cookie for many files)

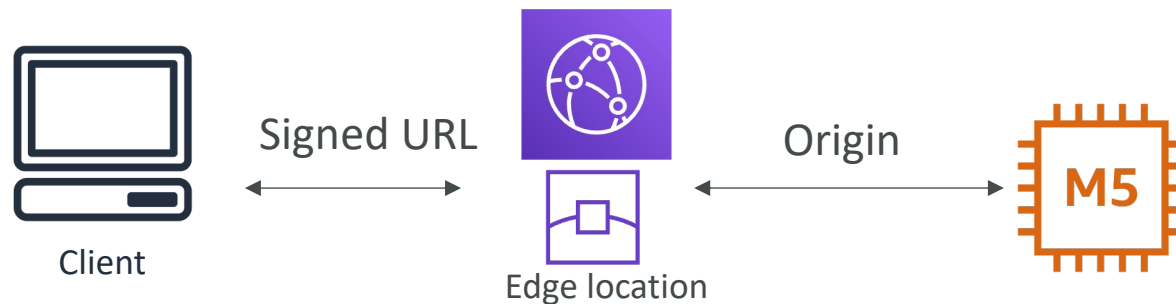
CloudFront Signed URL Diagram



CloudFront Signed URL vs S3 Pre-Signed URL

- CloudFront Signed URL:

- Allow access to a path, no matter the origin
- Account wide key-pair, only the root can manage it
- Can filter by IP, path, date, expiration
- Can leverage caching features



- S3 Pre-Signed URL:

- Issue a request as the person who pre-signed the URL
- Uses the IAM key of the signing IAM principal
- Limited lifetime

